

Applied Prelim

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1 Initial Data Processing and Visualization

1.a

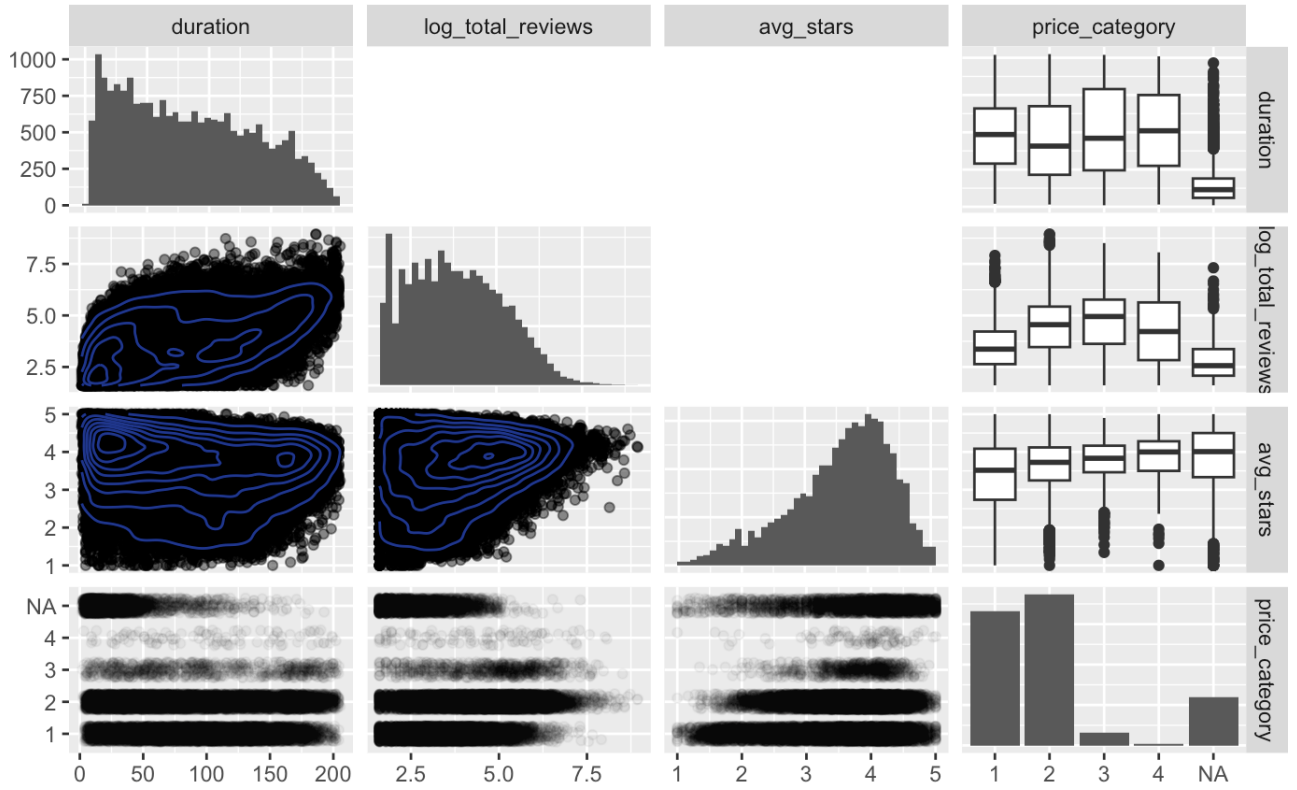
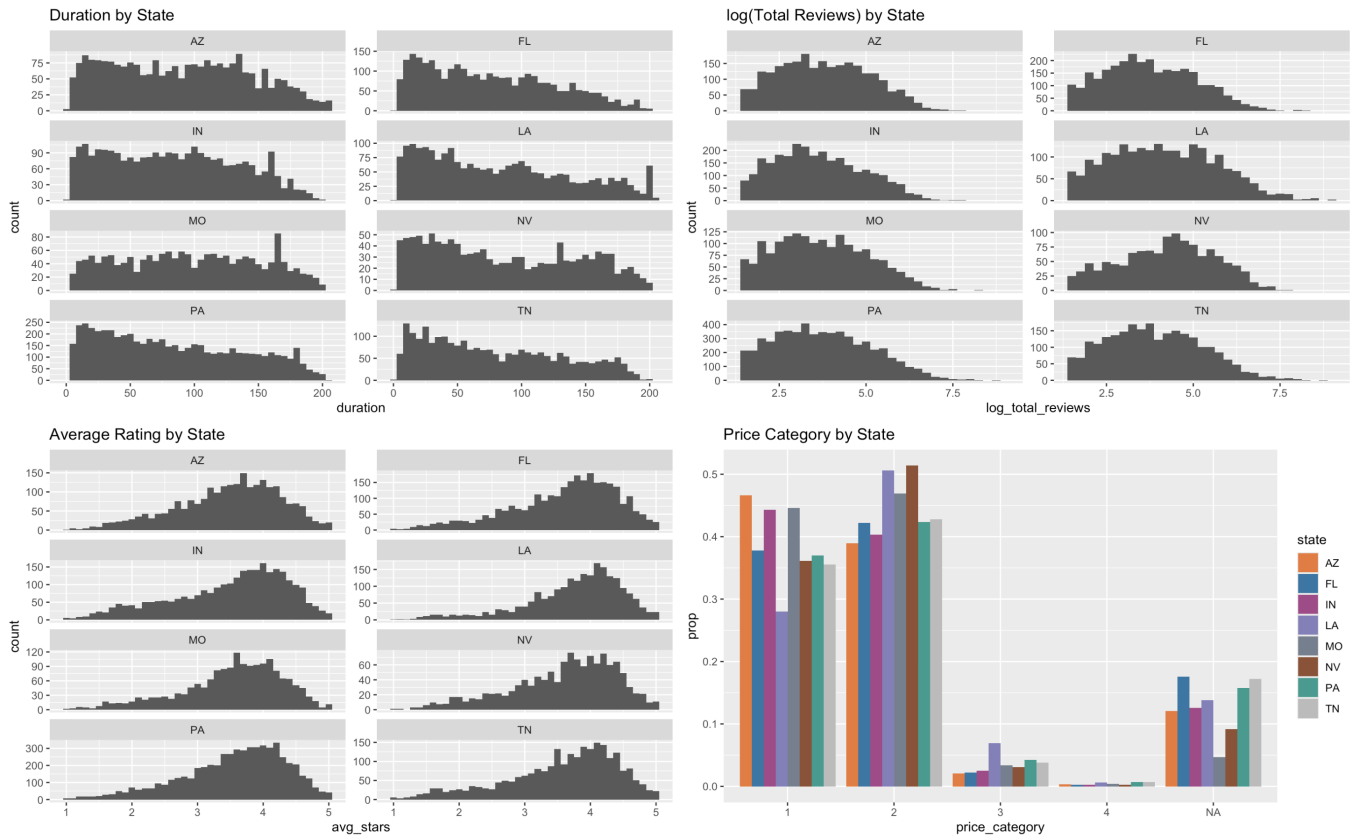


Figure 1: Histogram and Pairwise scatter plots with contour lines

We observe a few trends in the data while plotting:

- Total reviews is log transformed as it is much more informative. We will note this for appropriate transformations later.
- We see clear positive correlation between duration&total reviews& average stars.
- We observe the vast majority of restaurants are in price categories 1 or 2 with very few in 3 or 4.

1.b Plots By State



We observe a few patterns here:

- LA appears to have fewer cheap restaurants and more expensive ones relative the other states
- LA also appears to have the higher average stars with a weaker left tail than the rest
- Duration differs the most by state with some such as PA,FL,TN decaying as duration increases and others, such as NV,MO having a more flat distribution of restaurant duration.

2 Average Star Rating Across States

2.a

At a high level, the unweighted average can be understood as the average rating of restaurants in the state. The weighted average can be understood as the average dining experience that reviews reported. The first method treats restaurants as the units we are taking the average of and is a metric of average restaurant quality. In the second metric, because we are weighting by number of reviews, is an average of the reviews from the state.

2.b Unweighted

In order to estimate the standard error of each of these metrics. I did not assume the variance in constant across all restaurants because in reality some restaurants are more polarizing(as we will see in question 4) and therefore a pooled variance un-reasonable so I treated them separately. I do assume that the different reviews are independent which i deemed a reasonable since they are presumably made by different people at different times.

$$\begin{aligned}\bar{Y} &= \frac{1}{N} \sum_{i=1}^n \bar{Y}^{(i)} = \frac{1}{N} \sum_{i=1}^N \frac{1}{n_i} \sum_{j=1}^{n_i} y_j^{(i)} \\ \text{Var}(\bar{Y}) &= \frac{1}{N^2} \sum_{i=1}^N \frac{1}{n_i^2} \sum_{j=1}^{n_i} \text{Var}(y_j^{(i)}) \\ &= \frac{1}{N^2} \sum_{i=1}^N \frac{\sigma_i^2}{n_i}\end{aligned}$$

I estimate the variance for each restaurant σ_i using the standard sample variance estimate and compute the 95% CIs using normal quantiles.

State	Average	95% CI	99.375% CI
AZ	3.468	(3.458 , 3.477)	(3.453 , 3.482)
FL	3.595	(3.586 , 3.603)	(3.581 , 3.608)
IN	3.481	(3.472 , 3.490)	(3.467 , 3.495)
LA	3.748	(3.740 , 3.757)	(3.735 , 3.762)
MO	3.521	(3.510 , 3.532)	(3.504 , 3.538)
NV	3.586	(3.575 , 3.597)	(3.569 , 3.603)
PA	3.558	(3.551 , 3.564)	(3.548 , 3.567)
TN	3.566	(3.557 , 3.575)	(3.553 , 3.580)

Table 1: Unweighted Average Stars

In order to test if the highest Average Rating(LA) is statistically significant we can compare the confidence intervals and see if the lower bound of LA is above the upper bound of any other variable. However, in order to correct for the multiple testing issue with 8 CIs we need to compute $(1-\alpha/8)\%$ CIs in order to have a joint 95%CI. In doing so we still see that LA has significant evidence of the highest unweighted mean rating.

2.c Weighted

The process here is similar to that of part b:

$$\begin{aligned}\bar{Y} &= \frac{1}{\sum_{i=1}^n n_j} \sum_{i=1}^n n_j \bar{Y}^{(i)} = \frac{1}{\sum_{i=1}^n n_j} \sum_{i=1}^N \sum_{j=1}^{n_i} y_j^{(i)} \\ \text{Var}(\bar{Y}) &= \frac{1}{(\sum_{i=1}^n n_j)^2} \sum_{i=1}^N \sum_{j=1}^{n_i} \text{Var}(y_j^{(i)}) \\ &= \frac{1}{(\sum_{i=1}^n n_j)^2} \sum_{i=1}^N \sum_{j=1}^{n_i} \sigma_i^2 \\ &= \frac{\sum_{i=1}^N n_i \sigma_i^2}{(\sum_{i=1}^n n_j)^2}\end{aligned}$$

We see that LA also has significant evidence of the highest weighted mean rating.

State	Average	95% CI	99.375% CI
AZ	3.737	(3.732 , 3.741)	(3.730 , 3.743)
FL	3.846	(3.843 , 3.850)	(3.841 , 3.852)
IN	3.858	(3.854 , 3.862)	(3.852 , 3.864)
LA	3.949	(3.946 , 3.952)	(3.944 , 3.953)
MO	3.853	(3.849 , 3.858)	(3.846 , 3.860)
NV	3.790	(3.785 , 3.795)	(3.783 , 3.797)
PA	3.812	(3.810 , 3.815)	(3.809 , 3.816)
TN	3.869	(3.865 , 3.872)	(3.863 , 3.874)

Table 2: Weighted Average Stars

2.d Difference

State	AZ	FL	IN	LA	MO	NV	PA	TN
Diff	0.2692859	0.2519466	0.3769254	0.2002466	0.3324527	0.2040300	0.2544591	0.3023170

Table 3: Weighted Average Stars - UnWeighted Average Stars

We do see a statistically significant difference where the weighted procedure has higher averages across all states(as seen by comparing confidence intervals again). This makes sense because we observed positive correlation between number of ratings and average rating so we expect the weighting to draw the average up. One potential explanation is that restaurants with high rating are better or more popular and are thus expected to have more visitors and therefore more ratings than ones with lower ratings.

We also saw the confidence intervals are much tighter for the weighted procedure. This is because mean rating with more samples have lower variance and therefore giving them higher weights which results in a lower standard error.

3

3.a

As a first step I plotted a simple linear model with all covariates of interest and plotted the residuals.

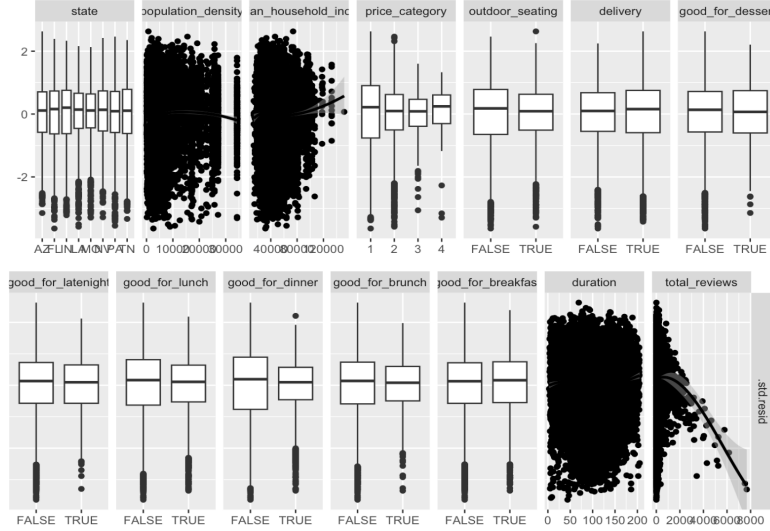


Figure 2: Residual plots for base Linear Model

We see clear issues with both heteroskedasticity and non-linearity. A first step is a transformation of the covariates and both Total Reviews and Median household income have improved fit under a log transform.

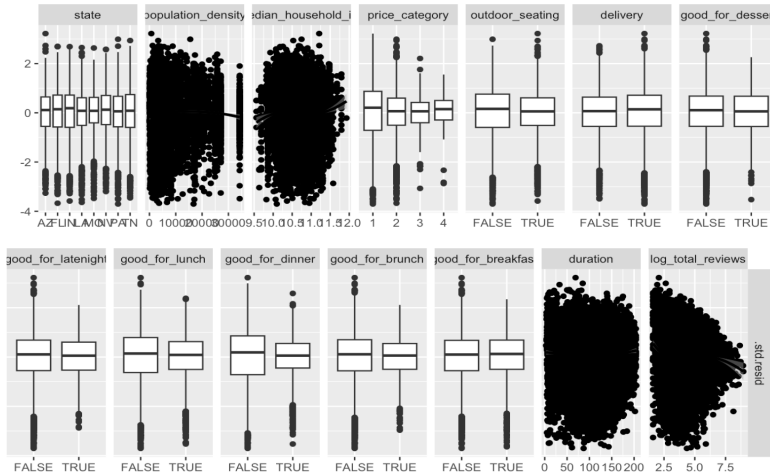


Figure 3: Residual plots for base Linear Model with log review, log income

I next perform a backwards step wise regression using BIC as the criteria to select the covariates of interest and then use stepwise regression to add interactions of those covariates.

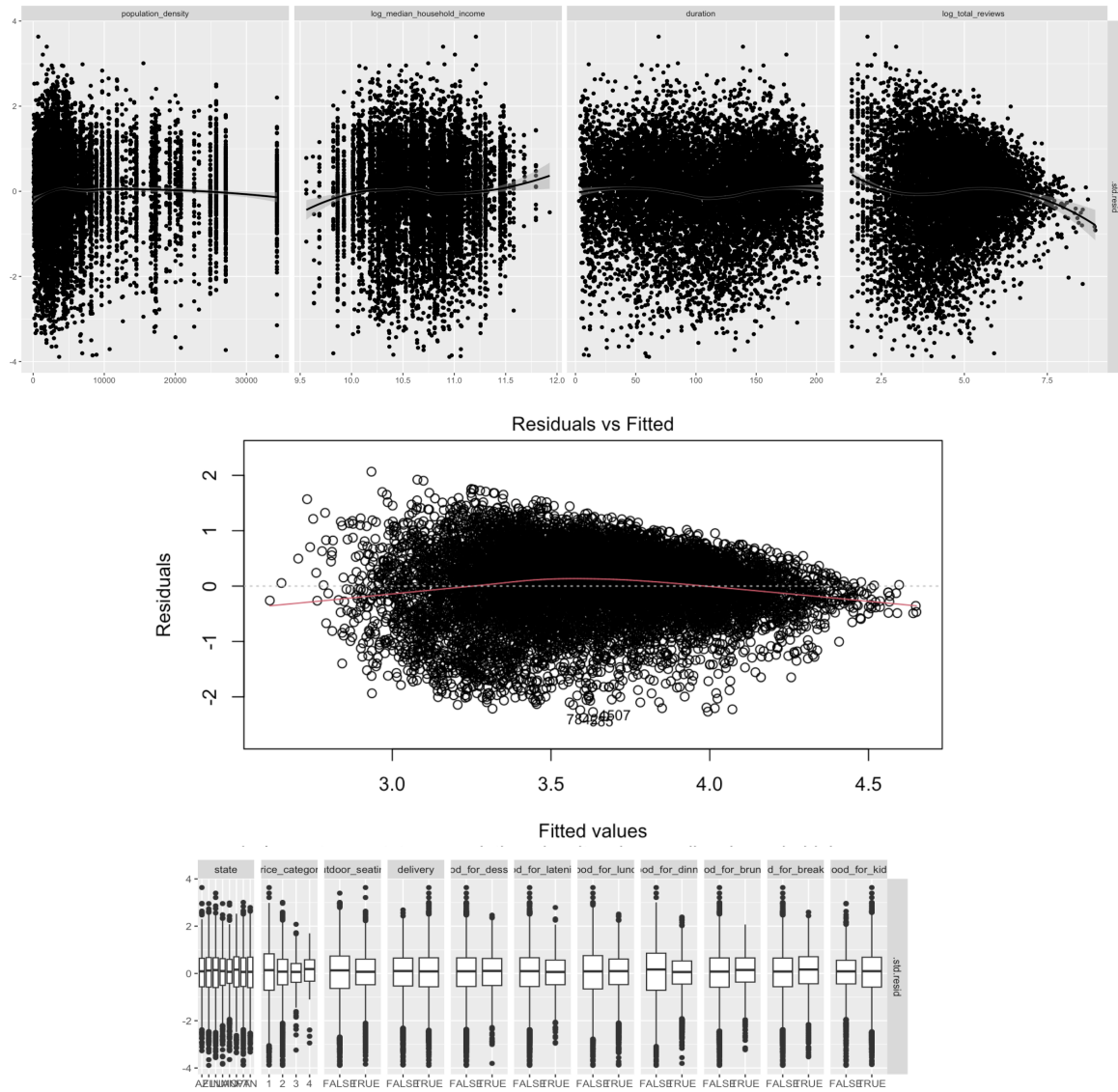


Figure 4: Residual plots for model with interactions

The resulting has improved linearity but still has non-linear trends in the residuals and heteroskedasticity with respect to the number of reviews so the inference results below based on this model may be questionable.

In our model we find that state, population density, log median household income, price category, outdoor seating, delivery, dessert, latenight, lunch, kids, duration and log total reviews all have an impact on the average stars.

```
lm(formula = avg_stars ~ state + population_density + log_median_household_income +
  price_category + outdoor_seating + delivery + good_for_dessert +
  good_for_latenight + good_for_lunch + good_for_kids + duration +
  log_total_reviews + price_category:delivery + price_category:good_for_lunch +
  duration:log_total_reviews + delivery:good_for_lunch + outdoor_seating:log_total_reviews +
  population_density:log_total_reviews + good_for_dessert:duration +
  good_for_latenight:good_for_kids + delivery:duration + delivery:log_total_reviews +
  log_median_household_income:log_total_reviews + delivery:good_for_dessert +
  outdoor_seating:good_for_lunch + population_density:good_for_lunch +
  population_density:outdoor_seating, data = d)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.727e+00	6.865e-01	9.800	< 2e-16 ***
stateFL	8.771e-02	2.407e-02	3.643	0.000270 ***
stateIN	1.103e-01	2.400e-02	4.597	4.35e-06 ***
stateLA	1.106e-01	2.588e-02	4.272	1.96e-05 ***
stateMO	1.917e-01	2.826e-02	6.783	1.24e-11 ***
stateNV	-5.679e-02	2.919e-02	-1.945	0.051751 .
statePA	4.805e-02	3.037e-02	1.582	0.113583
stateTN	3.970e-02	2.518e-02	1.577	0.114866
population_density	2.404e-05	3.631e-06	6.620	3.77e-11 ***
log_median_household_income	-3.307e-01	6.489e-02	-5.096	3.53e-07 ***
price_category2	-1.048e-01	2.665e-02	-3.933	8.46e-05 ***
price_category3	1.631e-02	4.954e-02	0.329	0.742072
price_category4	7.429e-02	1.196e-01	0.621	0.534366
outdoor_seatingTRUE	4.097e-01	5.010e-02	8.179	3.19e-16 ***
deliveryTRUE	-5.917e-01	5.043e-02	-11.734	< 2e-16 ***
good_for_dessertTRUE	2.545e-01	6.009e-02	4.235	2.31e-05 ***
good_for_latenightTRUE	-9.412e-02	3.687e-02	-2.553	0.010696 *
good_for_lunchTRUE	2.763e-01	3.004e-02	9.197	< 2e-16 ***
good_for_kidsTRUE	-1.225e-02	2.011e-02	-0.609	0.542592
duration	-6.471e-03	5.025e-04	-12.877	< 2e-16 ***
log_total_reviews	-3.951e-01	1.437e-01	-2.749	0.005980 **
price_category2:deliveryTRUE	3.158e-01	2.882e-02	10.956	< 2e-16 ***
price_category3:deliveryTRUE	3.009e-01	7.042e-02	4.272	1.95e-05 ***
price_category4:deliveryTRUE	3.202e-01	1.880e-01	1.704	0.088467 .
price_category2:good_for_lunchTRUE	-2.883e-01	2.603e-02	-11.073	< 2e-16 ***
price_category3:good_for_lunchTRUE	-4.456e-01	9.964e-02	-4.472	7.83e-06 ***
price_category4:good_for_lunchTRUE	-6.405e-01	3.289e-01	-1.948	0.051479 .
duration:log_total_reviews	8.552e-04	1.001e-04	8.544	< 2e-16 ***
deliveryTRUE:good_for_lunchTRUE	1.773e-01	2.683e-02	6.609	4.05e-11 ***
outdoor_seatingTRUE:log_total_reviews	-5.996e-02	1.087e-02	-5.516	3.55e-08 ***
population_density:log_total_reviews	-3.128e-06	6.901e-07	-4.533	5.88e-06 ***
good_for_dessertTRUE:duration	-2.292e-03	5.342e-04	-4.291	1.79e-05 ***
good_for_latenightTRUE:good_for_kidsTRUE	-2.398e-01	4.799e-02	-4.996	5.94e-07 ***
deliveryTRUE:duration	-1.716e-03	2.652e-04	-6.471	1.02e-10 ***
deliveryTRUE:log_total_reviews	7.306e-02	1.311e-02	5.575	2.54e-08 ***
log_median_household_income:log_total_reviews	5.091e-02	1.358e-02	3.750	0.000178 ***
deliveryTRUE:good_for_dessertTRUE	-1.850e-01	5.323e-02	-3.476	0.000511 ***
outdoor_seatingTRUE:good_for_lunchTRUE	-9.359e-02	2.582e-02	-3.625	0.000291 ***
population_density:good_for_lunchTRUE	-5.669e-06	1.589e-06	-3.568	0.000362 ***
population_density:outdoor_seatingTRUE	-5.722e-06	1.621e-06	-3.530	0.000417 ***

Figure 5: Coefficients for model with interactions

3.b

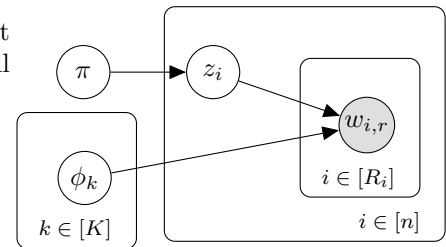
For this question, used a paired t-test to assess the difference in average rating between a restaurant's opening month and its second month. Since we are using average rating as our metric, I filtered the data to only include points where both the opening month and the month immediately have at least one review. I found evidence that average rating decreases from month 1 to month 2. There is an average decrease of roughly 0.155 points and a 95% CI of (0.1335034, 0.1774574). I also considered discarding restaurants with few ratings and got similar results.

4 EM

4.a Derivations

In EM we maximize the ELBO by alternating between optimizing w.r.t the Q and (ϕ, π) . I will derive the update equations and the algorithm will then loop through them until convergence.

$$ELBO = \mathbb{E}_{z \sim Q} \left[\log \left(\frac{P(\pi, \phi, w, z)}{Q(Z)} \right) \right]$$



4.a.1 M-Step

Conditioned on z , we can clearly see that π and ϕ are independent. Thus we can maximize them separately:

$$\begin{aligned} \pi &\leftarrow \arg_{\pi} \max \mathbb{E}_{z \sim Q} \left[\log \left(\frac{P(\pi, \phi, w, z)}{Q(Z)} \right) \right] \\ &= \arg_{\pi} \max \mathbb{E}_{z \sim Q} \left[\log \left(\frac{P(\pi | \phi, w, z) P(\phi, w, z)}{Q(Z)} \right) \right] \\ &= \arg_{\pi} \max \mathbb{E}_{z \sim Q} [\log (P(\pi | \phi, w, z))] && \text{terms irrelevant to } \pi \text{ dropped} \\ &= \arg_{\pi} \max \mathbb{E}_{z \sim Q} [\log (P(\pi | z))] && \pi \text{ of everything conditioned on } z \\ &= \arg_{\pi} \max \mathbb{E}_{z \sim Q} \left[\log \left(\text{Dirichlet} \left(\alpha_0 + \sum_{i=1}^n z_i \right) \right) \right] && \text{Dirichlet Multinomial Conjugacy} \\ &= \arg_{\pi} \max \mathbb{E}_{z \sim Q} \exp \left[\log \left(\text{Dirichlet} \left(\alpha_0 + \sum_{i=1}^n z_i \right) \right) \right] \\ &= \arg_{\pi} \max \text{Dirichlet} \left(\alpha_0 + \sum_{i=1}^n \mathbb{E}_{z \sim Q} [z_i] \right) \\ \pi_k &\leftarrow \frac{\alpha_k - 1}{\sum \alpha_k - K}, \quad \alpha_k = \alpha_0 + \sum_{i=1}^n \mathbb{E}_{z \sim Q} [\mathbb{1}_{z_i=k}] \end{aligned}$$

Since $\alpha \geq \alpha_0 > 1$ Dirichlet Mode has closed form

The second last equality holds because the geometric mean of an exponential family is the is mean of its natural parameters. We proceed similarly for ϕ :

$$\begin{aligned} \phi &\leftarrow \arg_{\phi} \max \mathbb{E}_{z \sim Q} \left[\log \left(\frac{P(\pi, \phi, w, z)}{Q(Z)} \right) \right] \\ &= \arg_{\phi} \max \mathbb{E}_{z \sim Q} [\log (P(\phi, w, z))] && \text{similar to prior} \\ &= \arg_{\phi} \max \mathbb{E}_{z \sim Q} \left[\log \left(\underbrace{P(\phi | z)}_{=P(\phi)} \prod_{i \in [n]} \prod_{r \in [R_i]} P(w_{ir} | \phi_k, z_i = k) \right) \right] && \text{conditions on } z_i = k, P(w_{ir}) \text{ only depends on } \phi_k \end{aligned}$$

Now, because $P(\phi)$ are sampled independently from each other the above factorizes and we can optimize them independently

$$\begin{aligned}
\phi_k &\leftarrow \arg_{\phi_k} \max \mathbb{E}_{z \sim Q} \left[\log \left(P(\phi_k) \prod_{i \in [n]} \mathbb{1}_{z_i=k} \prod_{r \in [R_i]} P(w_{ir} | \phi_k) \right) \right] \\
&= \arg_{\phi_k} \max \mathbb{E}_{z \sim Q} \left[\log \left(\text{Dirichlet} \left(\alpha_0 + \sum_{i \in [n]} \mathbb{1}_{z_i=k} \sum_{r \in [R_i]} w_{ir} \right) \right) \right] \quad \text{Dirichlet Multinomial Conjugacy} \\
&= \arg_{\phi_k} \max \text{Dirichlet} \left(\alpha_0 + \sum_{i \in [n]} \mathbb{E}_{z \sim Q} [\mathbb{1}_{z_i=k}] \sum_{r \in [R_i]} w_{ir} \right) \\
\phi_k^{(s)} &\leftarrow \frac{\alpha_k^s - 1}{\sum \alpha_k^s - 5}, \quad \alpha_k^s = \alpha_0 + \sum_{i \in [n]} \mathbb{E}_{z \sim Q} [\mathbb{1}_{z_i=k}] \sum_{r \in [R_i]} \mathbb{1}_{w_{ir}=s}
\end{aligned}$$

4.a.2 E-Step

For the E-step we update Q to maximize the ELBO. We know this occur when Q is the posterior:

$$\begin{aligned}
P(z_i = k | \pi, \phi, w) &= \frac{P(z_i = k, w | \pi, \phi)}{\sum_{k' \in [K]} P(z_i = k', w | \pi, \phi)} \\
&= \frac{P(z_i = k | \pi) P(w_i | z_i = k, \phi)}{\sum_{k' \in [K]} P(z_i = k' | \pi) P(w_i | z_i = k', \phi)} \\
&= \frac{\pi_k \prod_{s=1}^5 (\phi_k^s)^{\sum_{r \in [R_i]} \mathbb{1}_{w_{ir}=s}}}{\sum_{k' \in [K]} \pi_{k'} \prod_{s=1}^5 (\phi_{k'}^s)^{\sum_{r \in [R_i]} \mathbb{1}_{w_{ir}=s}}}
\end{aligned}$$

For Numerical stability this is computed in log space using:

$$\log \left(\pi_k \prod_{s=1}^5 (\phi_k^s)^{\sum_{r \in [R_i]} \mathbb{1}_{w_{ir}=s}} \right) = \log(\pi_k) + \sum_{s=1}^5 \sum_{r \in [R_i]} (\mathbb{1}_{w_{ir}=s}) \log(\phi_k^s)$$

For EM, we loop though these 3 update methods until convergence

4.b

The update equation as derived above only depend on $w_{i,r}$ through $\sum_{r \in [R_i]} \mathbb{1}_{w_{ir}=s}$. This quantity is the number of s star reviews for restaurant i which is that data that we are given. To understand why this occurs we note that the model as specified has no features to differentiate different reviews from each other (other than the number of stars). This means that only the number of reviews of each star are need to fit this model. To state this formally, we only need the sums because $P(w_{ir} = s | (\sum_{r \in [R_i]} \mathbb{1}_{w_{ir}=1}, \dots, \sum_{r \in [R_i]} \mathbb{1}_{w_{ir}=5}))$ is independent of all parameters in our model.

4.c

See appendix for code

4.d

For this part I will refer to the matrix of review counts as $(W_{is} \leftarrow \sum_{r \in [R_i]} \mathbb{1}_{w_{ir}=s})$. Creating a train test split by assigning reviews to train with probability p is equivalent to

$$W_{is}^{(\text{train})} \sim \text{Binom}(p, W_{is}), \quad W_{is}^{(\text{test})} = W_{is} - W_{is}^{(\text{train})}$$

I implemented this split and trained the models with the ascribed K . In order to assess performance I use the metric:

$$ERR = \frac{1}{5s} \sum_{i \in [n]} \sum_{s \in [5]} \left(\hat{W}_{is} - W_{is}^{(\text{test})} \right)^2$$

$$\hat{W}_{is} = R_i^{(\text{test})} \sum_{k=1}^K \phi_k^{(s)} \mathbb{E}_{z \sim Q} [\mathbb{1}_{z_i=k}]$$

$\hat{W}_{is}$ is the expected value of $\hat{W}_{is}$ under the posterior predictive distribution of the model. When the model is fitting well we expect the posterior predictive mean to be close to the test data. I plotted this error versus K and 15 latent components was selected based on the plot.

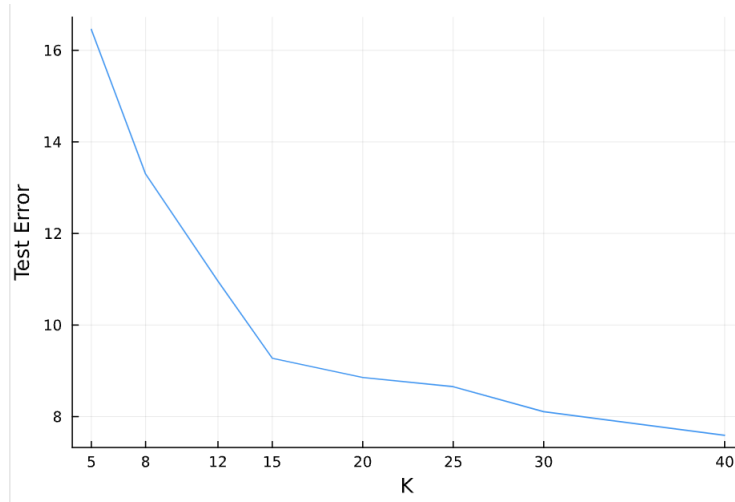


Figure 6: $\frac{1}{5s} \sum_{i \in [n]} \sum_{s \in [5]} \left(\hat{W}_{is} - W_{is}^{(\text{test})} \right)^2$ vs K

4.e Exploratory analysis

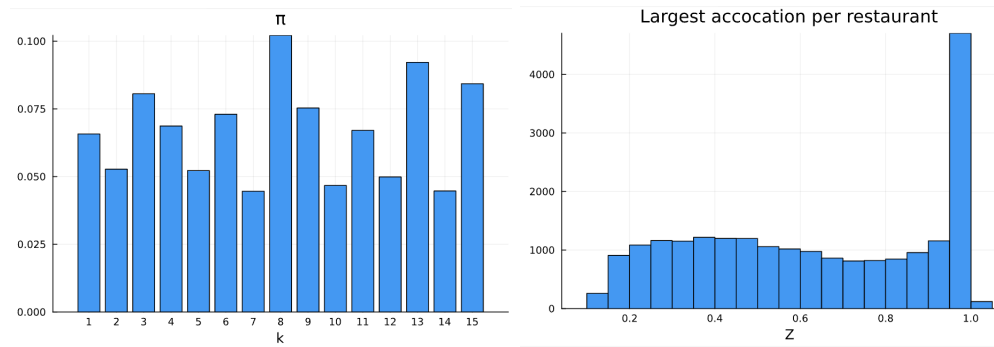


Figure 7: Values of π_k (left) and maximum z_{ik} for per restaurant i (right)

Plotting the π we see that all of the latents have fairly high activation. The most heavily weighted are classes 8,13,15. We will look at 8 and 15 in a little more detail below as they have interesting patterns. We also see that many restaurants have a spike in their latent allocation and belong almost entirely to one component.

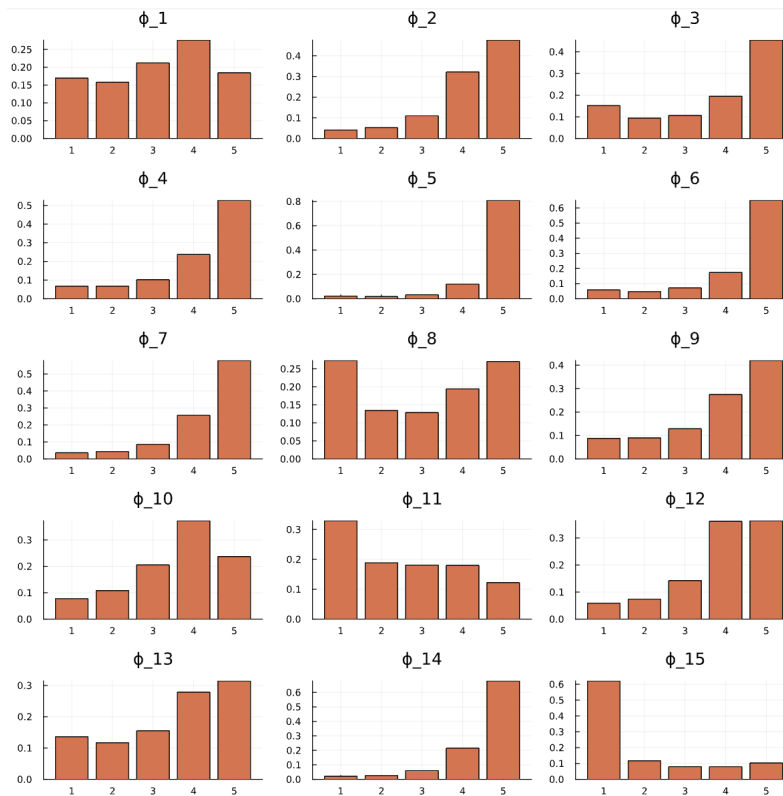


Figure 8: Values of each latent competent

Looking briefly at the component 8 we see a class that corresponds to the hypothesized class of restaurants that are highly polarizing(similar trends exist in components 3 and 13 though not as strongly. Latent classes 15 and 5 are also of interest as the have predominantly 5 and 1 star review respectively and investigating the members of these classes may reveal trends in what is

rated very negatively or very positively. Component 15 in particular is very different than any other components and I looked at the restaurants that were highly activated in this component ($z_{15} > 0.9$).

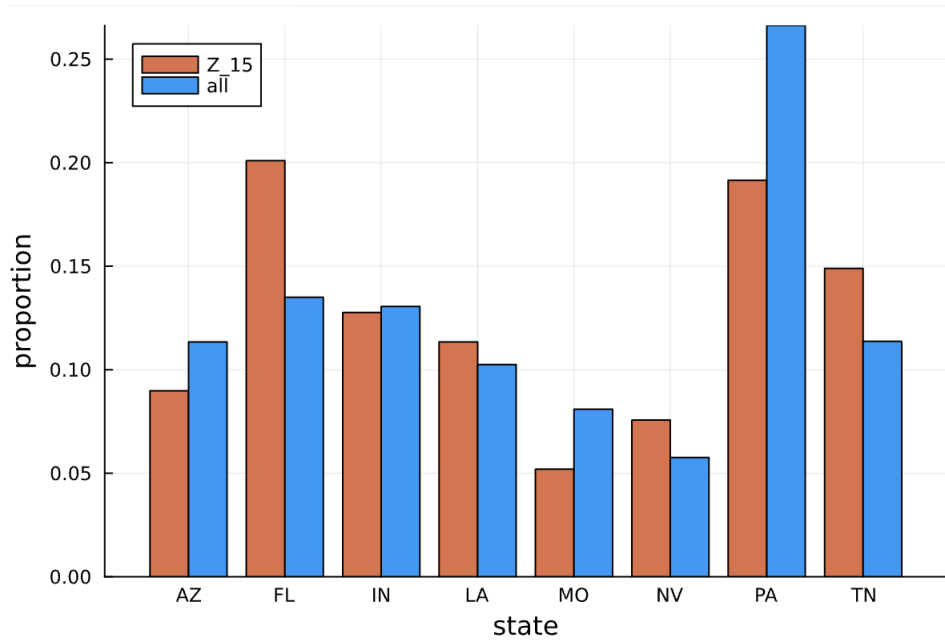


Figure 9: Distribution of states over all restaurants and over class 15 ($z_{15} > 0.9$)

One interesting trend within this class of restaurants is the distribution of states. We see that Florida is over greatly represented in this class of very poorly reviewed restaurants and Pennsylvania is greatly under represented.

5 Popularity via frequency of reviews

In order to model the rate of reviews, a natural model is a Poisson model of the number of reviews with an offset term of $\log(\text{duration})$. I fit an initial Poisson GLM as described with all of restaurant features as covariates.

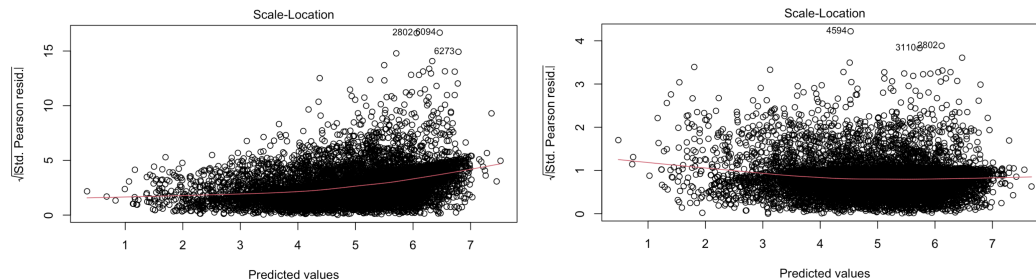


Figure 10: Residual plot for Poisson GLM(Left), Negative Binomial GLM(Right)

We see clear evidence of overdispersion in the Poisson GLM so I consider and Negative Binomial model which is much improved and we will proceed using that model.

I then perform backwards step wise regression based on BIC to select the covariates of interest and consider interaction of those selected covariates using step wise regression again.

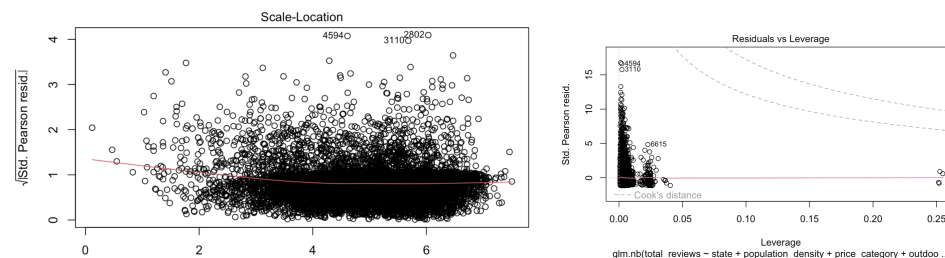


Figure 11: Diagnostic Plots for Negative Binomial GLM with interactions

We see 4 high leverage points in the Negative Binomial GLM with interactions. In investigating them further they all of price category 4, and 3 of the 4 are 'good for lunch' and 2 are 'good for dinner'. In the entire datasets, there are only 50 points in highest price category, 4 of which that are good for lunch, 3 of which that are good for dinner. Thus leaving there high leverage points in seems like the correct course. To verify that these points should be left, I removed them and re-fit. The significant changes to the fit were only in the interaction with price category and good for lunch. Since the rest of fit is not significantly changes we leave these high leverage points in but note that interaction with price category 4 and good for lunch may not be reliable as they are based on few restaurants.

```
glm.nb(formula = total_reviews ~ state + population_density +
  price_category + outdoor_seating + good_for_dessert + good_for_lunch +
  good_for_dinner + good_for_brunch + good_for_breakfast +
  good_for_dinner:good_for_breakfast + good_for_lunch:good_for_dinner +
  state:population_density + outdoor_seating:good_for_lunch +
  price_category:good_for_lunch + good_for_lunch:good_for_brunch +
  population_density:good_for_dessert + offset(log(duration)),
  data = d, init.theta = 1.48444283, link = log)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.187e+00	5.512e-02	-21.531	< 2e-16 ***
stateFL	1.280e-01	8.620e-02	1.485	0.137508
stateIN	-1.935e-01	8.214e-02	-2.356	0.018468 *
stateLA	-8.462e-02	1.006e-01	-0.841	0.400454
stateMO	-1.918e-01	8.165e-02	-2.349	0.018816 *
stateNV	3.445e-01	7.232e-02	4.763	1.91e-06 ***
statePA	-1.603e-01	6.503e-02	-2.464	0.013731 *
stateTN	1.519e-01	6.773e-02	2.242	0.024933 *
population_density	6.021e-05	1.368e-05	4.403	1.07e-05 ***
price_category2	9.045e-01	3.039e-02	29.766	< 2e-16 ***
price_category3	1.001e+00	5.476e-02	18.287	< 2e-16 ***
price_category4	1.196e+00	1.259e-01	9.504	< 2e-16 ***
outdoor_seatingTRUE	4.037e-01	2.541e-02	15.889	< 2e-16 ***
good_for_dessertTRUE	2.594e-01	4.610e-02	5.627	1.84e-08 ***
good_for_lunchTRUE	5.833e-01	3.207e-02	18.188	< 2e-16 ***
good_for_dinnerTRUE	6.739e-01	3.167e-02	21.280	< 2e-16 ***
good_for_brunchTRUE	5.704e-01	4.818e-02	11.840	< 2e-16 ***
good_for_breakfastTRUE	3.332e-01	3.351e-02	9.946	< 2e-16 ***
good_for_dinnerTRUE:good_for_breakfastTRUE	-5.907e-01	6.278e-02	-9.410	< 2e-16 ***
good_for_lunchTRUE:good_for_dinnerTRUE	-2.406e-01	4.044e-02	-5.950	2.68e-09 ***
stateFL:population_density	1.899e-06	2.245e-05	0.085	0.932586
stateIN:population_density	7.264e-05	2.523e-05	2.880	0.003983 **
stateLA:population_density	5.588e-05	1.882e-05	2.970	0.002977 **
stateMO:population_density	-3.518e-07	1.793e-05	-0.020	0.984349
stateNV:population_density	4.321e-06	1.967e-05	0.220	0.826115
statePA:population_density	-3.053e-05	1.381e-05	-2.211	0.027046 *
stateTN:population_density	5.801e-05	1.903e-05	3.048	0.002304 **
outdoor_seatingTRUE:good_for_lunchTRUE	-1.548e-01	3.345e-02	-4.628	3.69e-06 ***
price_category2:good_for_lunchTRUE	-1.823e-01	3.828e-02	-4.762	1.91e-06 ***
price_category3:good_for_lunchTRUE	9.650e-02	1.364e-01	0.707	0.479331
price_category4:good_for_lunchTRUE	-1.701e+00	4.322e-01	-3.936	8.29e-05 ***
good_for_lunchTRUE:good_for_brunchTRUE	-2.006e-01	5.633e-02	-3.561	0.000369 ***
population_density:good_for_dessertTRUE	1.714e-05	5.286e-06	3.242	0.001188 **

Figure 12: Residual plot for Poisson GLM(Left), Negative Binomial GLM(Right)

Based on the fitted coefficients we see that all of the included coefficients for good for — are positive where as all the coefficients for interaction between two of these such terms are negative. Based on this I would advise that a restaurant will focus on one meal rather than generalizing and that. Additionally outdoor seating has a positive effect on review rate as does increased population density, although the amount of benefit varies by state.

There are a number of issues with this analysis. The biggest issue is using rate of reviews as a proxy for popularity. There can be many reasons to leave a review, both positive and negative, and just looking at the number of reviews without looking at if they are positive or negative may be misleading.

Additionally we have no data on restaurants that have actually made changes. It may not be the case that changing a restaurant is the same as having a new restaurant with different qualities.

A Code

A.1 Code for Q1

A.1.a

```

1 # {r}
2 library(tidyverse)
3 library(ggify)
4 library(GGally)
5 library(MASS)
6 library(xtable)
7 select <- dplyr::select
8 # {r}
9
10 # {r}
11 data = read_csv('yelp.csv')
12 data = data %>% mutate(year_month = as_date(year_month, format = "%m/%d/%Y"))
13 # {r}
14
15 # {r}
16 #note: Verified that the sum(monthly_stars_N) = the max(total_stars_N) for all ID except #NAME?
17 #note: Also Verified that the sum(total_stars_N) = total_reviews
18 #note: Also Verified that the reported average star number is unweighted and matches manual computation with rounding error
19 #note Also Verified that the restaurant properties do not change for a restaurant
20 data %>% filter(business_id != "#NAME?") %>%
21   group_by(business_id) %>%
22     summarise(earliest_month = min(year_month),
23               latest_month = max(year_month),
24               duration = latest_month - earliest_month,
25               total_stars_1 = max(total_stars_1),
26               total_stars_2 = max(total_stars_2),
27               total_stars_3 = max(total_stars_3),
28               total_stars_4 = max(total_stars_4),
29               total_stars_5 = max(total_stars_5),
30               mtotal_stars_1 = sum(monthly_stars_1),
31               mtotal_stars_2 = sum(monthly_stars_2),
32               mtotal_stars_3 = sum(monthly_stars_3),
33               mtotal_stars_4 = sum(monthly_stars_4),
34               mtotal_stars_5 = sum(monthly_stars_5),
35               total_reviews = max(total_reviews),
36               mtotal_reviews = sum(monthly_reviews),
37               avg_stars = total_stars[which(year_month == latest_month)] ,
38               manual_avg_stars = (mtotal_stars_1 +
39                                   2*mtotal_stars_2 +
40                                   3*mtotal_stars_3 +
41                                   4*mtotal_stars_4 +
42                                   5*mtotal_stars_5)/total_reviews
43     ) %>%
44     filter( total_reviews != mtotal_reviews |
45             total_stars_1 != mtotal_stars_1 |
46             total_stars_2 != mtotal_stars_2 |
47             total_stars_3 != mtotal_stars_3 |
48             total_stars_4 != mtotal_stars_4 |
49             total_stars_5 != mtotal_stars_5 |
50             total_reviews != total_stars_1 + total_stars_2 + total_stars_3 + total_stars_4 + total_stars_5 |
51             abs(avg_stars - manual_avg_stars) > 1e-9 )
52
53
54 data %>% filter(business_id != "#NAME?") %>%
55   group_by(business_id) %>%
56   summarise( across(2:19, n_distinct) ) %>% select(-business_id) %>% {colSums(. != 1)}
57 # {r}

```



```

65 ▸ ```{r}
66 pivoted_data = data %>% filter(business_id!="#NAME?") %>%
67   group_by(business_id) %>%
68   summarise(
69     across(2:19,last),
70     earliest_month = min(year_month),
71     latest_month = max(year_month),
72     duration = time_length(interval(earliest_month,latest_month),unit = "months")+1 ,
73     total_stars_1 = max(total_stars_1),
74     total_stars_2 = max(total_stars_2),
75     total_stars_3 = max(total_stars_3),
76     total_stars_4 = max(total_stars_4),
77     total_stars_5 = max(total_stars_5),
78     total_reviews = max(total_reviews),
79     log_total_reviews = log(total_reviews),
80     avg_stars = total_stars[which(year_month == latest_month)])
81 ▸ ```
82 |
83 ▸ ```{r}
84 my_scatter = function(data, mapping,...){
85   ggplot(data, mapping) +
86     geom_point(...) +
87     geom_density_2d(alpha = .5)
88 }
89 my_scatter3 = function(data, mapping,...){
90   ggplot(data, mapping) +
91     stat_binhex()
92 }
93 my_scatter2 = function(data, mapping,...){
94   ggplot(data, mapping) +
95     geom_jitter(height = 0.25, ...)
96 }
97 pivoted_data %>%
98   select(duration,log_total_reviews,avg_stars,price_category) %>%
99   mutate(price_category = factor(price_category)) %>%
100   ggpairs(lower = list(continuous = wrap(my_scatter,alpha = 0.5),combo = wrap(my_scatter2,alpha = 0.05)),
101     diag = list(continuous = wrap("barDiag",bins = 40)),
102     upper = list(continuous = "blank",combo = "box_no_facet")
103   )
104 ▸ ```

```

A.1.b

```

107 ▸ ```{r}
108 p1 = pivoted_data %>%
109   ggplot() +
110   geom_histogram(aes(x =duration),binwidth = 5)+
111   ggh4x::facet_wrap2( ~state, scales = "free_y",nrow = 4, ncol = 2)+
112   labs(title = "Duration by State")
113
114 p2 = pivoted_data %>%
115   ggplot() +
116   geom_histogram(aes(x = log_total_reviews),binwidth = .25)+
117   ggh4x::facet_wrap2( ~state, scales = "free_y",nrow = 4, ncol = 2)+
118   labs(title = "log(Total Reviews) by State")
119
120 p3 = pivoted_data %>%
121   ggplot() +
122   geom_histogram(aes(x =avg_stars ),binwidth = .1)+
123   ggh4x::facet_wrap2( ~state, scales = "free_y",nrow = 4, ncol = 2)+
124   labs(title = "Average Rating by State")
125
126 p4 = pivoted_data %>% mutate(price_category = factor(price_category)) %>%
127   count(price_category, state) %>%
128   group_by(state) %>%
129   mutate(prop = n / sum(n)) %>%
130   ggplot(aes(x = price_category, y = prop, fill = state)) +
131   geom_col(position = position_dodge()) +
132   scale_fill_grafify(palette = "vibrant")+labs(title = "Price Category by State")
133 library(cowplot)
134 plot_grid(p1, p2,p3,p4 )
135 ▸ ```

```

A.2 Code for Q2

```

140 * ``{r}
141 * my_var = function(avg,w1,w2,w3,w4,w5,t){
142   sum(( c(1,2,3,4,5)-avg )^2*c(w1,w2,w3,w4,w5) )/(t-1)
143 * }
144 my_var = Vectorize(my_var)
145
146 d1 = pivoted_data %>%
147   mutate(Var = my_var(pivoted_data$avg_stars,
148     total_stars_1,total_stars_2,total_stars_3,total_stars_4,total_stars_5,
149     total_reviews),
150
151     SS = my_SS(pivoted_data$avg_stars,
152     total_stars_1,total_stars_2,total_stars_3,total_stars_4,total_stars_5,
153     total_reviews)) %>% group_by(state )
154 d2 = d1%>%
155   summarise(num_restaurants = n(),
156     num_reviews = sum(total_reviews),
157     unweighted_avg = mean(avg_stars),
158     weighted_avg = weighted.mean(avg_stars, total_reviews),
159     unweighted_var = sum(Var/total_reviews)/num_restaurants^2,
160     weighted_var = weighted.mean(Var, total_reviews)/num_reviews)
161 d3 = d1%>% summarise(num_restaurants = n(),
162   num_reviews = sum(total_reviews),
163   unweighted_avg = mean(avg_stars),
164   pooled_var = weighted.mean(Var,(total_reviews-1)),
165   weighted_avg = weighted.mean(avg_stars, total_reviews),
166   unweighted_var = sum(pooled_var/total_reviews)/num_restaurants^2,
167   weighted_var = pooled_var/num_reviews)
168 a = .05/16
169 b = data.frame(State = d2$state,
170   `Unweighted Average` = d2$unweighted_avg,
171   `UnPooled-CI-2.5` = qnorm(a,mean = d2$unweighted_avg,sd = sqrt(d2$unweighted_var)),
172   `UnPooled-CI-97.5` = qnorm(1-a,mean = d2$unweighted_avg,sd = sqrt(d2$unweighted_var)),
173   `Pooled-CI-2.5` = qnorm(a,mean = d3$unweighted_avg,sd = sqrt(d3$unweighted_var)),
174   `Pooled-CI-97.5` = qnorm(1-a,mean = d3$unweighted_avg,sd = sqrt(d3$unweighted_var)))
175
176 c = data.frame(State = d2$state,
177   `weighted Average` = d2$weighted_avg,
178   `UnPooled-CI-2.5` = qnorm(a,mean = d2$weighted_avg,sd = sqrt(d2$weighted_var)),
179   `UnPooled-CI-97.5` = qnorm(1-a,mean = d2$weighted_avg,sd = sqrt(d2$weighted_var)),
180   `Pooled-CI-2.5` = qnorm(a,mean = d3$weighted_avg,sd = sqrt(d3$weighted_var)),
181   `Pooled-CI-97.5` = qnorm(1-a,mean = d3$weighted_avg,sd = sqrt(d3$weighted_var)))
182 b
183 c
184 d3
185
186 (b$UnPooled.CI.97.5 - b$UnPooled.CI.2.5) - (b$Pooled.CI.97.5 - b$Pooled.CI.2.5 )
187 (c$UnPooled.CI.97.5 - c$UnPooled.CI.2.5) - (c$Pooled.CI.97.5 - c$Pooled.CI.2.5 )
188 * ``

```

A.3 Code for Q3

A.3.a

```

190 ~ ```{r}
191 pivoted_data$log_median_household_income = log(pivoted_data$median_household_income)
192 pivoted_data$log_population_density = log(pivoted_data$population_density)
193 pivoted_data$log_duration = log(pivoted_data$duration)
194
195 vars = c( "state", "population_density", "log_median_household_income",
196 "price_category", "outdoor_seating", "delivery", "good_for_dessert", "good_for_latenight",
197 "good_for_lunch", "good_for_dinner", "good_for_brunch",
198 "good_for_breakfast", "good_for_kids", "duration", "log_total_reviews", "avg_stars")
199
200 pivoted_data[vars] %>% {colSums(is.na(.))}
201
202 d =pivoted_data[vars]%>% na.omit()%>%
203   mutate(price_category = factor(price_category),
204          state = factor(state))
205 n = d %>% nrow()
206
207
208 m1 = lm(avg_stars~. , data = d)
209 #summary(m1)
210 #autoplot(m1,which = 1:6)
211 #ggduo(augment(m1,d),vars[1:14],".std.resid")
212
213
214 #heteroskedasticity -> tranform median and total reviews
215 #still have non-linear trends to deal with
216 slm1 <- step(m1, k = log(n) ,trace=F)
217 summary(slm1)
218
219 #stepwise to pair features back
220 plot(slm1)
221 #ggduo(augment(slm1,d),vars[1:14],".std.resid")
222 summary(slm1)
223
224
225 slm2 <- step(slm1, k = log(n),
226             scope = avg_stars ~ (state + population_density + log_median_household_income +
227   price_category + outdoor_seating + delivery + good_for_dessert +
228   good_for_latenight + good_for_lunch + good_for_kids + duration+ log_total_reviews )^2,
229             direction = "f",
230             trace=F)
231 summary(slm2)
232 plot(slm2)
233 ggduo(augment(slm2,d),vars[c(1,4:13)],".std.resid")
234

```

A.3.b

```

273 `r{
274 d= data %>% filter(business_id!="NAME?") %>%
275   group_by(business_id) %>%
276   summarise(
277     first_month = min(year_month),
278     second_month = min(year_month[which(year_month != first_month)]),
279     duration = time_length(interval(first_month,second_month),unit = "months")+1 ,
280     first_month_avg = monthly_stars[which(year_month == first_month)],
281     second_month_avg = monthly_stars[which(year_month == second_month)],
282     first_month_num = monthly_reviews[which(year_month == first_month)],
283     second_month_num = monthly_reviews[which(year_month == second_month)],
284     last_month = is_last_month[which(year_month == second_month)],
285     months = n()
286   )
287
288 f = d %>% filter(d$first_month_num != 0 & d$second_month_num != 0 & duration ==2)
289 t.test(f$second_month_avg, f$first_month_avg,paired = TRUE )
290
291 g = d %>% filter(d$first_month_num >30 & d$second_month_num >30 & duration ==2 )
292 t.test(g$second_month_avg, g$first_month_avg,paired = TRUE )
293 qnorm(c(.025,.975),
294       mean = mean( f$second_month_avg - f$first_month_avg ) ,
295       sd = sd(f$second_month_avg - f$first_month_avg)/sqrt(length(f$second_month_avg) ))|
296 `r

```

A.4 Code for Q4

A.4.a EM Code

```

1  using ~/Documents/Applied/EM.jl
2  using DataFrames
3  using Random
4  using Distributions
5  using StatsBase
6  using StatsFuns: logsumexp
7  using Plots
8  using JLD2
9  using StatsPlots
10
11  #naming convention for variables: Name_LengthDim1_LengthDim2
12  #variable names match those used from the problem and in my derivations
13
14  #model struct containing hyperparameters that are set and will not change
15  struct Model
16      α₀::Float64 #hyperparam for dirichlets
17      K::Int #number of latent components
18      N::Int #number of restaurants
19      w_N_5::Array{Int, 2} #Number of each review by restaurants
20
21      Model(w_N_5,K,α₀ = 1.15) = new(α₀,K,size(w_N_5)[1],w_N_5)
22  end
23
24  mutable struct State
25      π_K::Array{Float64, 1}
26      φ_5_K::Array{Float64, 2}
27      z_K_N::Array{Float64, 2}
28      #inner constructor to randomly initialize based on the model
29      State(m::Model) = new( rand(Dirichlet(m.K, m.α₀)),
30                             rand(Dirichlet(5, m.α₀),m.K),
31                             rand(Dirichlet(m.K, m.α₀),m.N) #initialize beliefs randomly
32      )
33  end
34
35  #Wrapper to allow clean dispatch for Update functions
36  update!(x::Symbol,model,state) = update!(Val{x}(),model,state) #update! (generic function with 1 method)
37
38  #M-step update for π
39  update!(::Val{π_K},m::Model,s::State) = begin
40      q_K = sum(s.z_K_N,dims = 2)[:].+ m.α₀
41      s.π_K = mode(Dirichlet(q_K))
42  end #update! (generic function with 2 methods)
43
44  #M-step update for φ
45  update!(::Val{φ_5_K},m::Model,s::State) = begin
46      q_K_5 = s.z_K_N * m.w_N_5 .+ m.α₀
47      for k in 1:m.K
48          s.φ_5_K[:,k] = mode(Dirichlet(q_K_5[k,:]))
49      end
50  end #update! (generic function with 3 methods)
51
52  #E-step update for z
53  update!(::Val{z_K_N},m::Model,s::State) = begin
54      logp_K_N = log.(s.π_K) .+ (m.w_N_5 *log.(s.φ_5_K))'
55      for n in 1:m.N
56          s.z_K_N[:,n] = exp.(logp_K_N[:,n] .- logsumexp(logp_K_N[:,n]))
57      end
58  end #update! (generic function with 4 methods)
59
60  #run EM till converges
61  EM(m::Model, s::State,tol = 1e-10) = begin
62      for i in 1:5000
63          prev = deepcopy(s)
64          for var in fieldnames(State)
65              update!(var,m,s)
66          end
67          if maximum(abs.(prev.z_K_N- s.z_K_N)) < tol
68              @show i
69              break
70          end
71      end
72  end # EM (generic function with 2 methods)

```

A.4.b Evaluation/K selection

```

74  train_test_split(data,p=0.7) = begin
75      train = similar(data)
76      for i in eachindex(data)
77          train[i] = rand(Binomial(data[i],p))
78      end
79      return (train, data.-train)
80  end
81
82  evaluate(model::Model,s::State,test) = begin
83      PPD = exp(sum(log.(s.ϕ_5_K * s.z_K_N) .* test')/sum(test))
84      R_N = sum(test,dims = 2)
85      W_hat = (s.ϕ_5_K * s.z_K_N)' .* R_N
86      ERR = mean( ( W_hat.- test).^2 )
87      return (PPD,ERR)
88  end
89
90  w_N_5 = Matrix(pivoted_data[:,[:total_stars_1,total_stars_2,total_stars_3,total_stars_4,total_stars_5]])
91  (train,test) = train_test_split(w_N_5) | ERROR: UndefVarError: `train_test_split` not defined
92
93  K = [5, 8, 12, 15, 20, 25, 30, 40] | 8-element Vector{Int64}:
94
95  Models = [] | Any[]
96
97  States = []
98  for k in K
99      m = Model(train,k)
100      s = State(m)
101      EM(m,s)
102      push!(Models,m)
103      push!(States,s)
104  end
105  PPD = []
106  ERR = []
107  for (m,s) in zip(Models,States)
108      (ppd,err) = evaluate(m,s,test)
109      push!(PPD,ppd)
110      push!(ERR,err)
111  end
112  plot(K,ERR,xaxis = "K",yaxis = "Test Error",xticks=K,legend = false)
113

```

A.4.c Exploratory analysis

```

116 #fit to full data
117 K = 15 | 15
118 m = Model(w_N_5,K)
119 s = State(m)
120 EM(m,s)
121 #plot of activations of pi
122 bar(1:K, s.pi_K;title = "pi",legend=false,xticks=1:K, xaxis = "k") | Plot{Plots.GRBackend() n=1}
123
124 #plot of activations of phi
125 plt = plot(layout=(5,3),title = "phi_K",legend = false) | Plot{Plots.GRBackend() n=0}
126 for k in 1:15
127     bar!(plt ,1:5 , s.phi_5_K[:,k], subplot=k, title = "phi_k",legend = false)
128 end | ✓
129 plot!(size=(1000,1000)) | Plot{Plots.GRBackend() n=30}
130 plt | Plot{Plots.GRBackend() n=30}
131
132 #plot max Z
133 m = maximum(s.z_K_N,dims = 1)[: ] | 21517-element Vector{Float64}:
134 histogram(m,title = "Largest accocation per restaurant",xaxis = "Z",legend = false) | Plot{Plots.GRBackend() n=1}
135
136 #investigate Z_15 with plots
137 pivoted_data[:,z5] = s.z_K_N[5,:] | 21517-element Vector{Float64}:
138 pivoted_data[:,z15] = s.z_K_N[15,:] | 21517-element Vector{Float64}:
139 pivoted_data[:,z8] = s.z_K_N[8,:] | 21517-element Vector{Float64}:
140 filtered = filter(row -> row.z5> 0.9, pivoted_data) | 423x33 DataFrame
141
142
143
144 c = ( "state", "good_for_kids") | ("state", "good_for_kids")
145 a = plotCol(Symbol(c[2])) | Plot{Plots.GRBackend() n=2}
146 a | Plot{Plots.GRBackend() n=2}
147 √ plotCol(col) = begin
148     x = pivoted_data[:,col]
149     map = countmap(x,ones(length(x))./length(x))
150     v = collect(values(map))
151     k = collect(keys(map))
152     g = fill("all",length(k))
153     c = fill(1,length(k))
154
155     x = vec(filtered[:,col])
156     map = countmap(x,ones(length(x))./length(x))
157     v = [v; collect(values(map)) ]
158     k = [k; collect(keys(map)) ]
159     g = [g; fill("Z_15",length(map)) ]
160     c = [c; fill(2,length(map)) ]
161     groupedbar(k, v, group = g,xaxis = string(col),yaxis = "proportion",color = c)
162 end | plotCol (generic function with 1 method)

```

A.5 Code for Q5

```

279 vars = c("state", "population_density", "log_median_household_income",
280 "price_category", "outdoor_seating", "delivery", "good_for_dessert", "good_for_latenight",
281 "good_for_lunch", "good_for_dinner", "good_for_brunch",
282 "good_for_breakfast", "good_for_kids", "duration", "total_reviews", "avg_stars")
283
284 pivoted_data[vars]
285 d = pivoted_data[vars] %>%
286   na.omit() %>%
287   mutate(price_category = factor(price_category),
288          state = factor(state))
289 n = d %>% nrow()
290
291 #plot poisson and NB to compare
292 p1 = glm(total_reviews ~ state + population_density + log_median_household_income +
293   price_category + outdoor_seating + delivery + good_for_dessert +
294   good_for_latenight + good_for_lunch + good_for_dinner + good_for_brunch +
295   good_for_breakfast + good_for_kids + offset(log(duration)),
296   family = poisson, data = d)
297
298 nb1 = glm.nb(total_reviews ~ state + population_density + log_median_household_income +
299   price_category + outdoor_seating + delivery + good_for_dessert +
300   good_for_latenight + good_for_lunch + good_for_dinner + good_for_brunch +
301   good_for_breakfast + good_for_kids + offset(log(duration)),
302   data = d)
303
304 #plot(p1)
305 #plot(nb1)
306 #select NB features
307 nb1_step <- step(nb1, k = log(n), trace=F)
308 #plot(nb1_step)
309 #summary(nb1_step)
310
311 #select NB interactions
312 nb1_interact <- step(nb1_step, k = log(n),
313   scope = total_reviews ~ (state + population_density +
314   price_category + outdoor_seating + good_for_dessert + good_for_lunch +
315   good_for_dinner + good_for_brunch + good_for_breakfast)^2 +
316   offset(log(duration)),
317   trace=F)
318 plot(nb1_interact)
319 summary(nb1_interact)
320
321 #investigate high leverage
322 high_leverage = sort(hatvalues(nb1_interact), decreasing=TRUE, index.return=TRUE)
323 high_leverage = sort.hix[high_leverage$x>.15]
324 d[high_leverage,]
325 d %>% count(price_category, good_for_lunch)
326 d %>% count(price_category, good_for_dessert)
327
328 nb2 = glm.nb(total_reviews ~ state + population_density + log_median_household_income +
329   price_category + outdoor_seating + delivery + good_for_dessert +
330   good_for_latenight + good_for_lunch + good_for_dinner + good_for_brunch +
331   good_for_breakfast + good_for_kids + offset(log(duration)),
332   data = d_no_lev)
333
334 nb2_step <- step(nb2, k = log(n), trace=F)
335
336 nb2_interact <- step(nb2_step, k = log(n),
337   scope = total_reviews ~ (state + population_density +
338   price_category + outdoor_seating + good_for_dessert + good_for_lunch +
339   good_for_dinner + good_for_brunch + good_for_breakfast)^2 +
340   offset(log(duration)),
341   trace=F)
342
343 #compare with and without high leverage
344 data.frame(nb2_interact$coefficients, nb1_interact$coefficients, diff = nb2_interact$coefficients - nb1_interact$coefficients)
345
346 plot(nb1_interact)
347 summary(nb1_interact)
348

```